hdoc day 31 by trishil chauhan

Aim

To write a menu-driven C program that performs operations on a two-dimensional array: sum of all matrix elements, sum of each row stored in a one-dimensional array, and transpose of the matrix.

C Program

```
#include <stdio.h>
#define MAX 20

int main(void) {
    int a[MAX][MAX], rowSum[MAX];
    int rows, cols, choice, i, j, total;

    printf("Enter number of rows and columns (1-20): ");
    if (scanf("%d %d", &rows, &cols) != 2 ||
        rows < 1 || rows > MAX || cols < 1 || cols > MAX) {
        printf("Invalid matrix dimensions.\n");
        return 1;
    }

    printf("Enter %d matrix elements:\n", rows * cols);
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            if (scanf("%d", &a[i][j]) != 1) {
                printf("Invalid element.\n");
                return 1;
            }
        }
    }

    do {
        printf("\n--- MATRIX OPERATIONS ---\n");
        printf("1. Sum of all elements\n");
        printf("2. Sum of each row (stored in array)\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }

        switch (choice) {
            case 1:
                total = 0;
                for (i = 0; i < rows; i++)
                    for (j = 0; j < cols; j++)
                        total += a[i][j];
                printf("Sum of all elements = %d\n", total);
                break;

            case 2:
                for (i = 0; i < rows; i++) {
                    rowSum[i] = 0;
                    for (j = 0; j < cols; j++)
                        rowSum[i] += a[i][j];
                }
```

```
                printf("Row sums stored in array: ");
                for (i = 0; i < rows; i++)
                    printf("%d ", rowSum[i]);
                printf("\n");
                break;

            case 3:
                printf("Transpose (%d x %d):\n", cols, rows);
                for (j = 0; j < cols; j++) {
                    for (i = 0; i < rows; i++)
                        printf("%d\t", a[i][j]);
                    printf("\n");
                }
                break;

            case 4:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 4);

    return 0;
}
```

Sample Input and Output

```
Enter number of rows and columns (1-20): 2 3
Enter 6 matrix elements:
1 2 3
4 5 6

Choice 1: Sum of all elements = 21
Choice 2: Row sums stored in array: 6 15
Choice 3: Transpose (3 x 2):
1    4
2    5
3    6
Choice 4: Exiting program.
```

Result

The menu-driven program successfully calculates the total sum, stores each row sum in an array, displays the transpose, and exits when selected.